

School of Computer Science and Information Technology
Lucerne University of Applied Sciences and Arts (Switzerland)

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subtitle

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Declaration

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- ☐ Public (Standard)
- ☐ Private

Declaration

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Place/Date, Signature _____

Expression of thanks and gratitude

Thanks to my family, relatives and firends for all the support given to finish this thesis.

Author Name, 2025

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Summary

this is this is complete garbage and still even more garbage

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Glossary

Cognitive computing A set of theories and techniques to let computers to mimic the mechanisms of the human brain. It provides the basis for the practical application of cognition and learning theories to computer systems with the use of soft computing methods..

Computing with words and perceptions A process allowing to perform computations on words, phrases, and prepositions drawn from a natural language, which describe perceptions of people towards different aspects of the context they are surrounded by. This is based on the fuzzy logic toolbox and allows to represent and perform operations on the meaning of words..

Convolutional neural network A class of neural networks commonly used for image analysis that is relying on convolution operations to extract features from data..

Fuzzy logic An extension of classical binary logic, where the truth value of propositions can not only be completely true or false, but also partially true and false to varying degree..

Perceptual computing A set of theories and techniques allowing computers to compute and reason with perceptions and imprecise data..

Acronyms

HSLU Lucerne University of Applied Sciences and Arts.

NN Neural Network.

List of AI Tools Used

Placeholder for the list of AI tools used during the creation of this thesis.

- Tool name, purpose, and extent of use.

List of Formulas

Placeholder for the list of formulas used in this thesis.

- Formula name/number — brief description — page reference.

Chapter 1

Problem, Research Question, Vision

1.1 Problem

Placeholder for the problem description. Describe the issue or challenge that motivates this thesis.

1.2 Research Question

Placeholder for the research questions.

RQ1. Placeholder for research question 1.

RQ2. Placeholder for research question 2.

1.3 Vision

Placeholder for the vision. Describe the intended outcome or goal of the work.

Chapter 2

State of the Art

2.1 Related Work

Placeholder for related work. Review existing research, technologies, or practices relevant to the problem.

2.2 Background

Placeholder for background information. Introduce foundational concepts and terminology used throughout the thesis.

Chapter 3

Ideas and Concepts

3.1 Proposed Approach

Placeholder for the proposed approach. Outline the main ideas and conceptual design of the solution.

3.2 Conceptual Design

Placeholder for the conceptual design. Describe the architecture or framework developed in this thesis.

Chapter 4

Methods

4.1 Research Methodology

Placeholder for the research methodology. Describe the scientific or engineering methods applied.

4.2 Data and Tools

Placeholder for data and tools. Describe datasets, software, and instruments used.

4.3 Experimental Setup

Placeholder for the experimental setup. Describe how experiments or studies were designed and conducted.

Chapter 5

Realization

5.1 Implementation

Placeholder for the implementation. Describe how the concepts and methods were put into practice.

5.2 Technical Details

Placeholder for technical details. Provide specifics about the system, algorithms, or artefacts created.

5.3 Challenges and Solutions

Placeholder for challenges and solutions. Discuss obstacles encountered during realization and how they were resolved.

Chapter 6

Validation and Evaluation

6.1 Validation

Placeholder for validation. Describe how the correctness of the solution was verified.

6.2 Evaluation

Placeholder for evaluation. Present results, metrics, and a discussion of findings.

6.3 Discussion

Placeholder for discussion. Interpret the results in the context of the research questions.

Chapter 7

Outlook

7.1 Conclusion

Placeholder for the conclusion. Summarize the contributions and answer the research questions.

7.2 Future Work

Placeholder for future work. Suggest directions for further research or development.

Bibliography

Appendix A

Appendix A: Supplementary Material

Placeholder for supplementary material. Include additional data, figures, tables, or code listings that support the main text.

Appendix B

Appendix B: AI Usage

Placeholder for AI usage documentation. List all AI tools used during the creation of this thesis and describe how they were used.